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Discrete Mathematics

3Week [24 February - 01 March]

Quiz 3

Started on	2024-03-06 20:28
State	Finished
Completed on	2024-03-06 21:58
Time taken	1 hour 30 mins
Grade	14.00 out of 15.00 (93%)

Question 1

Correct

Mark 1.00 out of 1.00

Suppose $A = \{2, 4, 6\}$, $B = \{2, 6\}$, $C = \{4, 6\}$ and $D = \{4, 6, 8\}$.

Determine which of these sets are subsets of which other of these sets and choose correct answers.

Select one or more:

- ☐ A is a subset of B
- ☒ B is a subset of A ✓
- ☐ A is a subset of C
- ☒ C is a subset of A ✓
- ☒ C is a subset of D ✓
- ☐ D is a subset of B

Your answer is correct.

The correct answers are: B is a subset of A , C is a subset of A , C is a subset of D

Question 2

Correct

Mark 2.00 out of 2.00

What is the cardinality of each of these sets?

- a. $\emptyset$ 0 ✓
- b. $\{\emptyset\}$ 1 ✓
- c. $\{\emptyset, \{\emptyset\}\}$ 2 ✓
- d. $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$ 3 ✓

Question 3
Complete
Mark 3.00 out of 3.00

Find $\mathcal{P}(\mathcal{P}(\{a\}))$ and $|\mathcal{P}(\mathcal{P}(\{a\}))|$.

Write your answers on any paper, take a picture and attach it below.

 3_5.jpg

Comment:

Question 4
Correct
Mark 1.00 out of 1.00

Let $A = \{a, b, c, d, e\}$ and $B = \{b, c, d, e, f, g, h\}$. Then

a. $A \cup B$ is $\{$ ☒ $\}$

b. $A \cap B$ is $\{$ ☒ $\}$

Question 5
Complete
Mark 2.00 out of 2.00

Find the sets A and B if $A - B = \{1, 5, 7, 8\}$.

$B - A = \{2, 10\}$ and $A \cap B = \{3, 6, 9\}$.

Write your solution, take a picture and attach it below.

 3_5.jpg

Comment:

Question 6
Correct
Mark 1.00 out of 1.00

Fill in the truth table for $A - B$.

A	B	$A - B$				
1	1	0 ☒				
1	0	1 ☒				
0	1	0 ☒				
0	0	0 ☒				

Question **7**

Complete

Mark 3.00 out of 3.00

Show that $(A - C) \cap (C - B) = \emptyset$.

Write your proof, take a picture and attach it below.

 7_8.jpg

Comment:

Question **8**

Complete

Mark 1.00 out of 2.00

Let $A_i = (0, i)$. Find

a. $\bigcup_{i \geq 1} A_i$

b. $\bigcap_{i \geq 1} A_i$

Write your solution, take a picture and attach it below.

 7_8.jpg

Comment: